QUALYXUS

Multimodal Air-Gapped Edge-AI Platform for Zero-Scrap Manufacturing

- **Startup Name:** Qualyxus
- **Founder & Lead Architect:** Utkarsh Sengar (B.Tech CSE, IIT Dharwad)

> **We solve** late-stage material scrap and unpredicted machine downtime **by providing** a 100% air-gapped multimodal edge-AI platform that derives physical vibration metrics directly from camera pixels, **to help** high-precision defense, aerospace, and automotive manufacturers **accomplish** zero-defect production and 100% data sovereignty.

Compelling Problem Statement

The Problem: The Invisible Structural Blindspot

High-precision manufacturing facilities suffer from two disconnected operational silos that burn millions in profit:

1. Legacy Machine Vision (Post-Mortem Sorting)

- Inspects surface defects **after** machining is completed.
- Catches scrap only after high-value titanium/alloys, CNC hours, and labor are ruined.
- Completely blind to internal machine mechanics and tool drift.

2. Internal Machine Telemetry (Blind to Outcomes)

- Monitors motor heat and spindle vibration to prevent total machine breakdown.
- Completely blind to final part tolerances.
- A CNC machine can run at 100% "normal" internal parameters while actively producing 100% dimensional scrap.

Target Customer Profile: Plant Heads, VPs of Quality, and Chief Maintenance Officers in High-Precision CNC Machining, Aerospace & Defense, Automotive Components, and EV Battery lines.

Market Research & Opportunity

TAM / SAM / SOM Market Sizing

$20B

TAM (Total Addressable Market)

~₹1,65,000 Crores — Global Smart Manufacturing AI & Predictive Maintenance Market.

$2.5B

SAM (Serviceable Addressable Market)

~₹20,000 Crores — High-Precision Machining, Defense, Aerospace, & EV Component Lines in India & APAC.

$150M

SOM (Serviceable Obtainable Market)

~₹1,200 Crores — 2,000 High-Precision Production Lines in Indian industrial corridors (Bengaluru, Pune, Belagavi, Hyderabad, NCR).

Market Tailwinds & Growth Drivers

India PLI Expansion

Driven by India's ₹1.97 Lakh Crore PLI incentives in EV, Defense, and Semiconductor manufacturing.

The 80% Legacy Fleet Opportunity

Over 80% of CNC machines operating in India are 5–15 years old, requiring low-cost, non-invasive smart retrofits rather than ₹50L+ machine replacements.

Solution Offered & Innovation Proposed

The Causal Bridge: Unifying Vision + Physics

Qualyxus bridges product defects directly to machine mechanical causes using a single camera setup.



[CAMERA FEED] ——▶ [BARE-METAL EDGE ENGINE] ——▶ [PLC INPUT REGISTER] (One Camera, (Custom C++/CUDA, (Zero Liability, Zero Hardwiring) Sub-8ms Latency) Instant Deployment)

Core Deep-Tech Innovations

1 Multimodal AI Fusion	2 Bare-Metal Edge Engine	3 100% Air-Gapped Security
Camera feeds capture surface defects and component geometry, while machine sensors capture internal telemetry—fused together by our edge AI engine."	Custom Triton/CUDA fused kernels running locally on sub-$500 edge hardware (NVIDIA Jetson) at <8ms latency.	Zero internet dependency, fulfilling strict Defense data sovereignty laws.

Product & Service Details

Features, User Experience & Regulatory Judo

"All-Faces Inspector" Hardware Module

Proprietary multi-camera chassis delivering 360-degree omni-directional surface inspection with glare-free optics.

Bypassing the 3-Year Safety Certification Lock

- **System Classification:** Engineered strictly as a **Deterministic Forensic Advisory System**.
- **Input Register Write Protocol:** Writes directly to the factory's existing Safety PLC **Input Register**—never the output coil.
- **Zero Customer Liability:** The plant's pre-certified Safety PLC retains sole authority over physical stop execution, bypassing 3-year ISO 13849 safety recertification delays.

On-Premise Touchscreen Dashboard

Intuitive shop-floor interface displaying live sub-pixel tool drift graphs, vibration telemetry, and instant scrap counter metrics.

Competition & Comparison Matrix

How Qualyxus Captures the Lion's Share

Operational Method	Legacy Vision (Cognex / Keyence)	AI Vision Competitors (SwitchOn)	Heavy Automation (Siemens / Bosch)	QUALYXUS PLATFORM PPTX+ 1
Operational Method	Post-Mortem Sorter (catches scrap late)	Visual-Only Defect Check	Heavy Physical Sensor Wiring	**Live Pre-Defect Interception**
Data Underpinnings	Surface Image Matching	Image Pattern Matching	Machine Telemetry Logs	**Pixel → Vibration Physics (ISO Grounded)**
Deployment Time	High CapEx ($10k+ Racks)	Cloud/Hybrid Retraining	Multi-Month Line Downtime	**4-Hour Non-Invasive Setup (Zero Wiring)**
Safety Compliance	N/A (Post-Line)	N/A (Visual Check)	3-Year ISO Recertification Lock	**Advisory Engine (PLC Input Register Write)**
Budget Captured	Quality Control Only	Quality Control Only	Maintenance Only	**Dual Budget: Quality + Maintenance**
Per-Line Revenue	₹20L – ₹30L	₹30L – ₹40L	₹50L+ (High CapEx)	**₹40L – ₹50L / Year**

Business Model & Unit Economics

Primary Revenue Model: Hybrid Hardware-Enabled SaaS

We focus on **one highly profitable, recurring subscription stream**:

1. One-Time Setup & Hardware Fee: ₹2.0 Lakhs – ₹2.5 Lakhs per line

- Covers sub-$500 NVIDIA Jetson edge node, industrial cameras, lighting array, and 4-hour setup.
- Kept under ₹2.5L to fit within routine plant operational maintenance allowances, bypassing corporate CapEx board approvals.

2. Recurring Annual Software License (SaaS): ₹30 Lakhs / year

Taps both **Quality Control** and **Predictive Maintenance** corporate budgets simultaneously.

Go-To-Market (GTM) Strategy

The 4-Step Customer Acquisition Flywheel

1

1. Advisory Wedge (Land)

Install in under 4 hours via Safety PLC Input Register write, bypassing safety liabilities and legal reviews.

2

2. 30-Day ROI Proof (Prove)

Validate ₹15L–₹25L in direct scrap reduction on 1 baseline production line.

3

3. Plant-Wide Expansion (Expand)

Roll out software across remaining 10–20 production lines in the client facility.

4

4. Channel Distribution (Scale)

Partner with CNC machine distributors and industrial automation integrators to bundle Qualyxus on new machine sales.

Traction and Milestones

Empirically Validated 2-Month Pilot Traction

Our initial software deployment has achieved direct financial and operational validation across marquee manufacturing facilities:

1. Rossell Techsys (Aerospace & Defense)

- **Defect Reduction: 0.8% absolute scrap reduction**
- **Direct Annualized Savings: ₹20 Lakhs**

2. Magellan Aerospace (Precision Aerospace Components)

- **Defect Reduction: 0.2% absolute scrap reduction**
- **Direct Annualized Savings: ₹15 Lakhs**

3. Rotary Connectors (Precision Connector Line)

- **Defect Reduction: 0.5% absolute scrap reduction**
- **Direct Annualized Savings: ₹25 Lakhs**

Total Verified Initial Software Impact: Combined Annualized Financial Value: ₹60 Lakhs in savings across 3 active deployment lines. **Core Takeaway:** Serves as concrete empirical proof of software accuracy, early market validation, and rapid payback feasibility for industrial clients.

Team & Advisory Details

Leadership, Cap Table & Strategic Advisors

Founder & Core Architect

Utkarsh Sengar (B.Tech CSE, IIT Dharwad)

- Single-handedly engineered the core bare-metal C++/CUDA engine, 0.34ms DSP spectral core, and ViT-Mamba hybrid model.
- *LinkedIn:* [**linkedin.com/in/utkarsh-sengar**]

Cap Table Status

100% Founder Owned (Utkarsh Sengar)

Strategic Advisory Board

Dr. Rakesh Kumar

VP R&D, Rossell Techsys (ex-Moog, ex-Indian Navy).

Shankar Lal

Discipline Chief, Cyient (21 years at HAL).

Ramesh Rana

Operations Lead, SIG SAUER.

QUALYXUS

The Operating System for Zero-Scrap Global Manufacturing

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